Atharva Chougule

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Education

Stanford University *Sept 2025 – Present*
MS in Computer Science (Systems specialization)

- GPA: **4.13/4**

Indian Institute of Technology Madras *July 2019 – May 2023*
Bachelor of Technology with Honors in Computer Science and Engineering

- GPA: **9.79/10** | Class Rank: **1**
- **Gold Medalist** for being the student with **highest GPA** in Computer Science Department.

Research Experience

Compiling Sparse Tensor Algrbra for GPUs and Multicore CPUs *Stanford University*
Stanford Compiler Lab | Guide: Prof. Fredrick Kjolstad *Aug 2025 - Present*

- Developed a **Partitioning** technique for automatic parallelization of Sparse Tensor Algebra kernels.
- Extended the **TACO** compilation framework with this technique in the **Nacho Compiler**.
- Generated Code that outperforms existing vendor libraries **PyTorch, CuSparse, Intel MKL**.

Code Generation for Distributed Graph Algorithms *IIT Madras*
Bachelors Thesis | Guide: Prof. Rupesh Nasre *Dec 2022 - May 2023*

- Architected and built a **compiler** for the **distributed systems backend** of **StarPlat** - a DSL for graph analytics capable of generating code for multiple target architectures (**GPUs, multi-core and distributed systems**).
- Developed a new **graph representation** by modifying the **CSR** representation for storing **distributed dynamic graphs**.
- Generated **MPI** code for graph algorithms like Triangle Counting, Page Rank, Betweeness Centrality and SSSP.

Evaluating Byzantine Fault Tolerance for a new model of distributed computing *IIT Madras*
Research Project | Guide: Prof. John Augustine *Aug 2022 - Nov 2022*

- Modeled a new framework for **distributed computing** where, unlike the traditional MPC model, **data is decoupled from the distributed network** and is stored in an external entity like the Cloud.
- Studied **Byzantine Fault Tolerance** in the new model and developed a **Conversion theorem** from classic **MPC** algorithms.

Publications

Partitioning Unstructured Sparse Tensor Algebra for Load-Balanced Parallel Execution Paper|Code
Atharva Chougule, Alexander J Root, Rubens Lacouture, Bobby Yan, Rohan Yadav, Fredrik Kjolstad
ASPLOS, 2027

Generating Morph Algorithms for Multiple Backends Paper|Code
Ashwina Kumar, Nibedita Behera, **Atharva Chougule**, Mohammed Shan, Rushabh Lalwani, Rupesh Nasre

Work Experience

Waymo LLC *Mountain View, CA*
Software Engineer Intern | ML Compilers/Architecture Team *June 2026 – Present*

- Built a **JAX PJRT** backend for **Waymo's custom hardware**, enabling **JIT** compilation and execution of arbitrary custom JAX models on the platform.
- **Extending JAX with inline assembly support** support to JAX, enabling **ML hardware architects and performance engineers** to combine high-level JAX with hand-optimized assembly kernels for **hardware-specific performance tuning**.

Rubrik, Inc *Bangalore, India June*
Software Engineer | Rubrik Security Cloud Private (RSC-P) - Platform Team *2023 - April 2025*

- Engineered a **distributed and horizontally scalable architecture** for Rubrik Security Cloud Private (RSC-P) using

Kubernetes. This resolved the compute bottleneck for RSC-P's job framework, increasing its supported scale to more than **5 times**.

- Developed a **state machine** in Golang for RSC-P's upgrade framework to make cluster upgrades robust and predictable.
- Built **defragmentation tools** for the **MySQL database** resulting in **storage reduction by more than 70%**.
- **Led a hackathon project** for evaluating different **Kubernetes distros** and migrating the RSC-P architecture to **Rancher**.

Software Engineer 2 | Rubrik Security Cloud Private (RSC-P) - Platform Team *April 2025 - July 2025*

- Led a project for hardening **security and data resilience** of RSC-P by **enforcing backups and automating disaster recovery**.
- Improved system stability by optimizing resource utilization and addressing security gaps resulting in more than **50% decrease in customer-reported issues.**

Rubrik, Inc *Bangalore, India*
Software Engineer Intern | Rubrik Cloud Vault(RCV) *May 2022 – July 2022*

- Designed microservices-based framework to sync customer cost metrics for **Rubrik Cloud Vault(RCV)**. Built a general framework to sync Cost data from **Azure Cost API** to Rubrik's Database and trigger alerts for unusual cost patterns.

Technical Skills

- **Languages:** C++, C, CUDA, Java, Go, Python, Scala, Ocaml
- **Libraries:** OpenMP, MPI, Pandas, PyTorch, JAX, Scikit-Learn, Numpy
- **Technologies:** Kubernetes, Docker, CUDA, GCP, Azure

Academic Projects

TensorDSL for rPi-QPU *Spring 2026 | Code*
Course Project: CS240LX - Advanced Systems Laboratory | Guide: Prof. Dawson Engler *Stanford University*

- Designed and implemented a **DSL for Array programming** for the **Broadcom VideoCore GPU** on **Rasperry Pi**.
- Built a Compiler to lower simple high-level Tensor Programs to **QASM assembly** .

Speaker-Microphone Audio Bootloader *Winter 2026 | Code*
Course Project: CS140E - Operating Systems Implementation | Guide: Prof. Dawson Engler *Stanford University*

- Implemented a **audio bootloader** for Raspberry Pi to bootload and run any programs on the Pi using audio.
- **Implemented custom drivers** for a speaker and microphone to get **maximum audio-bandwidth** for the bootloader.

Compiler for MacroJava *Autumn 2022 | Code*
Course Project: Compiler Design | Guide: Prof. Kartik Nagar *IIT Madras*

- Developed a compiler for a subset of Java, supporting **conditionals, loops, control sequences, classes, etc**.
- Implemented a lexical analyser and parser using **Flex** and **Bison**. Translated the generated AST to **microIR**, which was further translated to **miniRA for register allocation** and was finally translated to the **MIPS Assembly**.

Scholastic Achievements and Honors

- Awarded the **B. Ravichandran Memorial Prize** for being the student with **best academic performance** and **highest GPA** in the Computer Science and Engineering Department. *2023*
- Nominated for **Engineering Excellence Award** at Rubrik. *2024*
- Awarded the prestigious **Kishore Vaigyanik Protsahan Yojana (KVPY)** fellowship *2019*

Teaching

CS143: Compilers *Spring 2026*
Teaching Assistant under Fredrik Kjolstad *Stanford University*

- Developed coursework, ran office hours, graded assignments, and managed course logistics for the Compilers course.

Mentoring

- **Mentored an intern** at Rubrik by providing him technical guidance and **helping him** on the project of Horizontal Pod Autoscaling (HPA) and Smaller Spec Support for Rubrik Security Cloud Private (RSC-P) through his internship.
- **Mentored new grads** transition from academic to industry setting by participating in the **Rubrik Mentorship Program**.